Linear Algebra Problems

UTSA Problem Solving Club

"The art of doing mathematics consists in finding that special case which contains all the germs of generality." — *David Hilbert*

The grazing cows

In 4 weeks, 12 cows graze bare $3\frac{1}{3}$ acres of pasture land, and in 9 weeks, 21 cows graze bare 10 acres of pasture land. Accounting for the uniform growth rate of grass and assuming equal quantities of grass per acre when the pastures are put into use, how many cows will it take to graze bare 24 acres of pasture land in a period of 18 weeks?

Source: Isaac Newton's Arithmetica Universalis

The Commuting Converse

Let A and B be $n \times n$ matrices with real entries such that:

$$A + B = AB$$

Prove that $AB = BA$.

Hint: Factor $AB - A - B + I$.

Source: Putnam Preparation

Skew-symmetry? That's odd!

Assume that n is odd, and let A be an $n \times n$ real matrix satisfying $A^T = -A$. Prove or disprove: *Claim:* A is singular.

Source:

Cubic-to-Quadratic

Let $A \neq B$ be real $n \times n$ matrices such that $A^3 = B^3$ and $A^2B = B^2A$. Can $A^2 + B^2$ be invertible?

Source: Putnam 1991 A2

Let S be a set of 2×2 integer matrices each of whose entries a_{ij} is a perfect (integer) square not exceeding 200. Show that if S has more than 50,387 elements, then it has two elements that commute.

Hint: $50{,}387 = 15^4 - 15^2 - 15 + 2$.

Source: Putnam 1990 B3

Additive translates in $\mathrm{GL}_2(\mathbb{Z})$

Let A, B be 2×2 matrices with integer entries such that A, $A+B$, $A+2B$, $A+3B$ and $A+4B$ are invertible whose inverses have integer entries. Show that $A+5B$ is also invertible whose inverse has integer entries.

Source: Putnam 1994 A4

Circle-in-Tesseract

What is the largest radius of a circle fitting inside a unit tesseract? (A tesseract is a 4-dimensional hypercube.)

Source: Putnam 2008 B3

Diverging GCDs

For $n \geq 1$, let d_n be the greatest common divisor of the entries of $A^n - I$, where I is the 2×2 identity matrix, and

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$$

Show that $\lim_{n\to\infty} d_n = \infty$.

Source: Putnam 1994 B4

Let S_1, S_2, $\ldots$, S_{2^n-1} be the nonempty subsets of $\{1, 2, \ldots, n\}$ in some order, and let M be the $(2n-1) \times (2n11)$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset \\ 1 & \text{otherwise.} \end{cases}$$

Calculate the determinant of M.

Source: Putnam 2018 A2

GOE_2

Let

$$A = \begin{bmatrix} X & Y \\ Y & Z \end{bmatrix}$$

be a random matrices whose entries are (obtained from the values of) three independent normal random variables X, Y, Z such that $\mathrm{Var}(X) = 1 = \mathrm{Var}(Z)$ but $\mathrm{Var}(Y) = 1/2$. Is it more likely that both eigenvalues of A have the same sign, or that they have different signs?

Source: Random Matrix Theory